



AI-Powered Hypertension Health Education & Monitoring Platform

Project Documentation

🏠 Nabd — Project Documentation

1. Introduction

1.1 Project Name

Nabd - AI-Powered Hypertension Health Education & Monitoring Platform

1.2 Project Description

Nabd is an Arabic-first digital health platform designed to help people with hypertension understand their condition, monitor their blood pressure readings, and access reliable educational information.

The platform combines:

- Artificial Intelligence
- Medical Knowledge Base
- Blood Pressure Monitoring
- Health Education
- User Authentication
- REST APIs
- Secure Data Storage

The main objective is to provide an easy-to-use platform that supports health awareness while maintaining strict safety boundaries around medical information.

2. Problem Statement

Hypertension is a common chronic condition, but many people face difficulties understanding their blood pressure readings and accessing reliable health information.

Common challenges include:

- Difficulty understanding blood pressure values.
- Lack of awareness about warning signs.
- Difficulty tracking measurements over time.
- Searching for medical information from unreliable sources.
- Difficulty understanding treatment and lifestyle recommendations.
- Lack of Arabic-first digital health education tools.

Nabd aims to address these problems through a centralized platform that combines educational content, AI assistance, and personal monitoring.

3. Project Objectives

The main objectives of Nabd are:

1. Provide reliable Arabic health education.
2. Help users record and monitor blood pressure readings.
3. Provide an AI assistant for general hypertension-related questions.
4. Ground AI responses in a predefined medical knowledge base.
5. Reduce unsupported or unsafe medical responses.
6. Protect users' personal health data.
7. Provide a scalable REST API.
8. Create a simple and accessible Arabic RTL interface.

4. Target Users

Patients / Users

People who want to:

- Learn more about hypertension.
- Record blood pressure readings.
- Review their previous measurements.
- Ask general health education questions.
- Understand lifestyle recommendations.

Future Users

Future versions could support:

- Doctors.
- Healthcare professionals.
- Clinics.
- Healthcare organizations.

5. Main System Features

5.1 User Authentication

Users can create accounts and log in securely. Authentication is handled through Supabase Authentication. The system maintains authenticated sessions and protects user-specific resources.

5.2 User Profile

Authenticated users can access their profile and update supported personal information, such as name, email, and basic profile information.

6. Blood Pressure Monitoring

Users can add blood pressure measurements. Each measurement may contain:

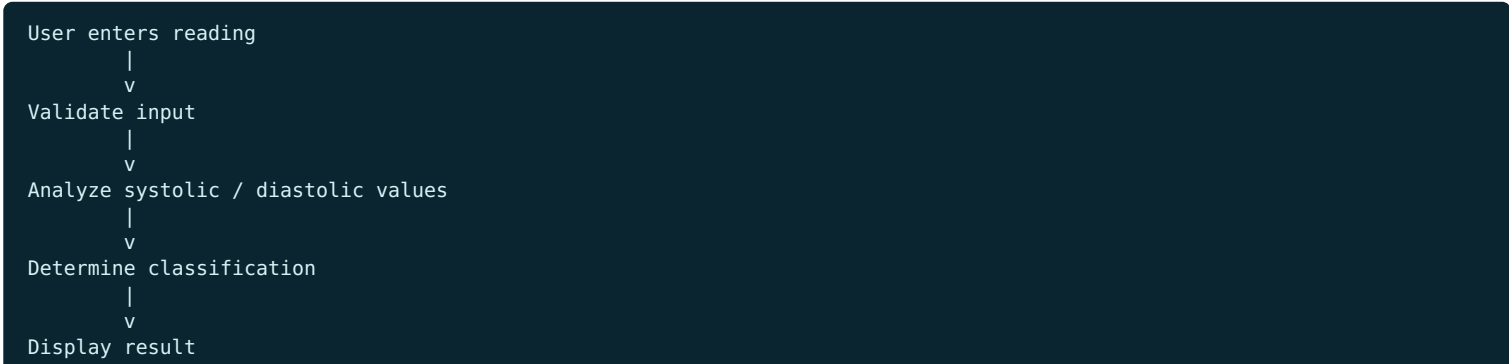
- Systolic pressure
- Diastolic pressure
- Pulse
- Notes
- Measurement date

The system validates the submitted values and stores them in the user’s account.

Users can: add a reading, view readings, view individual readings, update readings, delete readings, and view statistics.

7. Blood Pressure Classification

The application provides an educational classification of blood pressure readings.



The application may classify readings into categories such as: Normal / acceptable, Elevated, High, Danger, Low. The classification is intended for educational awareness and does not constitute a medical diagnosis.

8. AI Health Assistant

8.1 Overview

Nabd includes an AI-powered assistant that allows users to ask questions related to hypertension, such as:

- What is blood pressure?
- How can I measure my blood pressure correctly?
- What lifestyle changes can help?
- What are common medication side effects?
- When should I seek urgent medical attention?

9. AI Architecture

The AI system follows a knowledge-grounded architecture.



10. Knowledge Base

The project contains a structured medical knowledge base, stored in `src/data/kb/`.

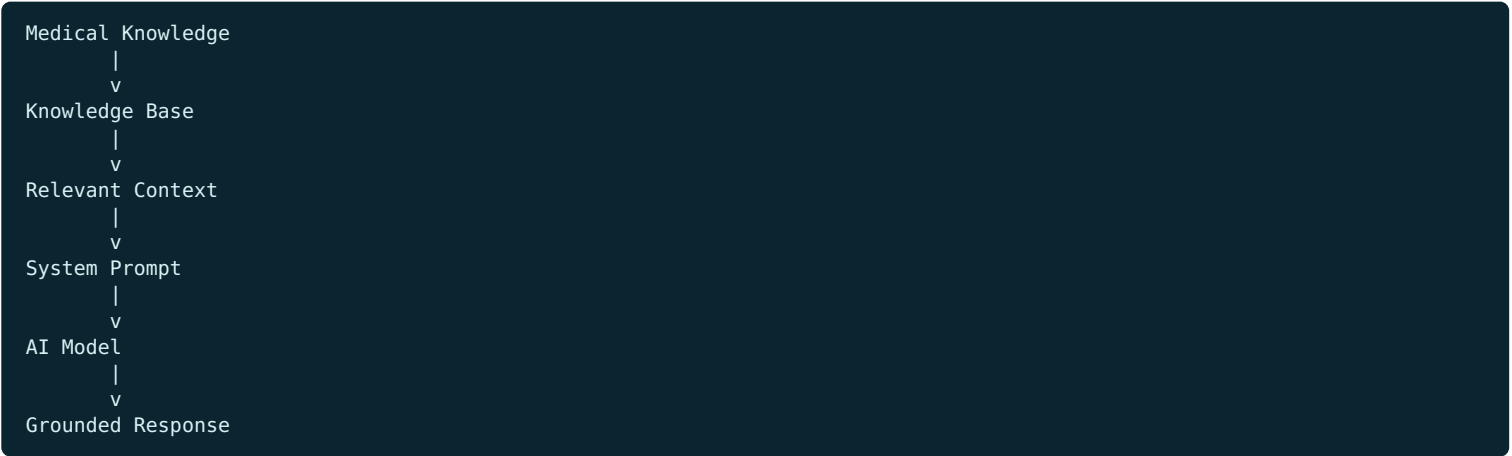
Current topics include:

- 01-starting-treatment.md
- 02-tests-and-followup.md
- 03-side-effects.md
- 04-lifestyle.md
- 05-measurement-and-emergencies.md

The content is based on trusted medical guidance, including the WHO 2021 guideline for pharmacological treatment of hypertension in adults.

11. RAG / Knowledge Grounding Approach

The current implementation uses a lightweight knowledge-grounding approach. Instead of allowing the AI model to answer entirely from its general training knowledge, the application provides predefined medical knowledge as context.



This approach helps keep the assistant focused on the project’s approved medical knowledge.

12. Future RAG Architecture

A more advanced version of Nabd can implement a complete Retrieval-Augmented Generation pipeline.



Potential technologies include: embedding models, PostgreSQL + pgvector, vector databases, semantic search, metadata filtering, reranking. This would allow the system to retrieve only the most relevant medical information for each user question.

13. AI Safety Layer

Because Nabd operates in a health-related domain, the AI assistant follows strict safety rules.

The assistant should not:

- Diagnose diseases.
- Prescribe medication.
- Change medication dosage.
- Recommend stopping medication.
- Replace a healthcare professional.
- Provide unsupported medical claims.

Instead, it should provide general educational information and recommend professional medical care when appropriate.

14. Emergency Detection

The system includes safety handling for potentially serious symptoms, such as:

- Chest pain or pressure
- Sudden weakness or numbness
- Sudden speech problems
- Sudden vision problems
- Severe shortness of breath
- Loss of consciousness
- Confusion
- Very high blood pressure readings with concerning symptoms

When such information is detected, the assistant prioritizes seeking urgent professional medical care rather than continuing a normal educational conversation.

15. Technology Stack

Frontend: React, TypeScript, TanStack Start, TanStack Router, Tailwind CSS, Radix UI, Lucide React, Recharts

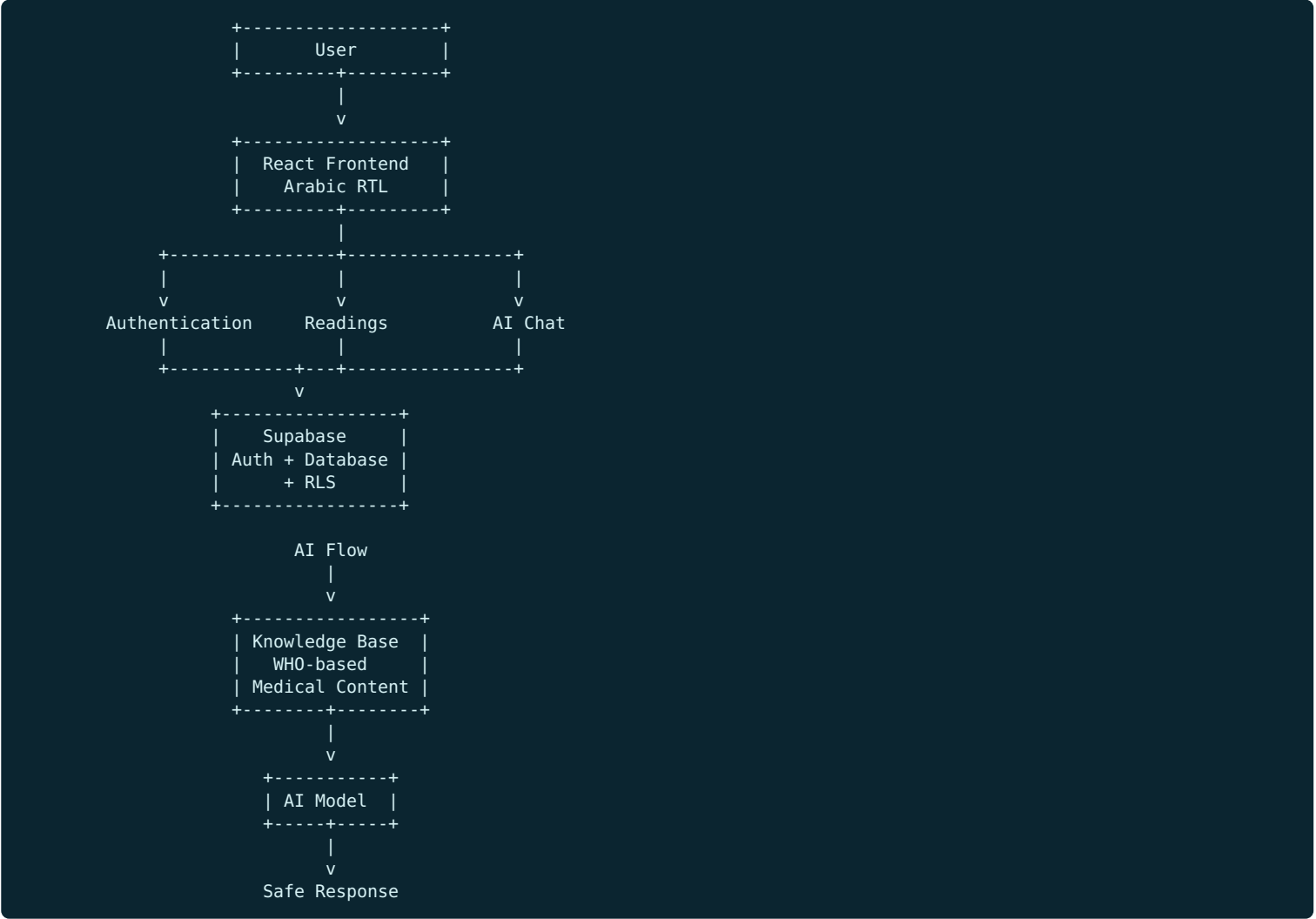
Backend: TanStack Start, Nitro, TypeScript, REST API

Database: Supabase, PostgreSQL

Authentication: Supabase Authentication

Artificial Intelligence: AI SDK, Gemini, Lovable AI Gateway, Knowledge-grounded prompting

16. System Architecture



17. Database

The application uses Supabase PostgreSQL as its database layer. The database stores application information such as user profiles, blood pressure readings, and user-related application data.

18. Row Level Security

Row Level Security (RLS) is used to protect user-specific data.

```
User A -> Can access -> User A's data only
User B -> Can access -> User B's data only
```

This prevents one authenticated user from accessing another user’s private measurements.

19. REST API

The application exposes a REST API under `/api/public/v1`.

Public Endpoints

```
GET /health
GET /articles
GET /articles/:slug
POST /chat
```

Protected Endpoints

```
GET /profile
PATCH /profile

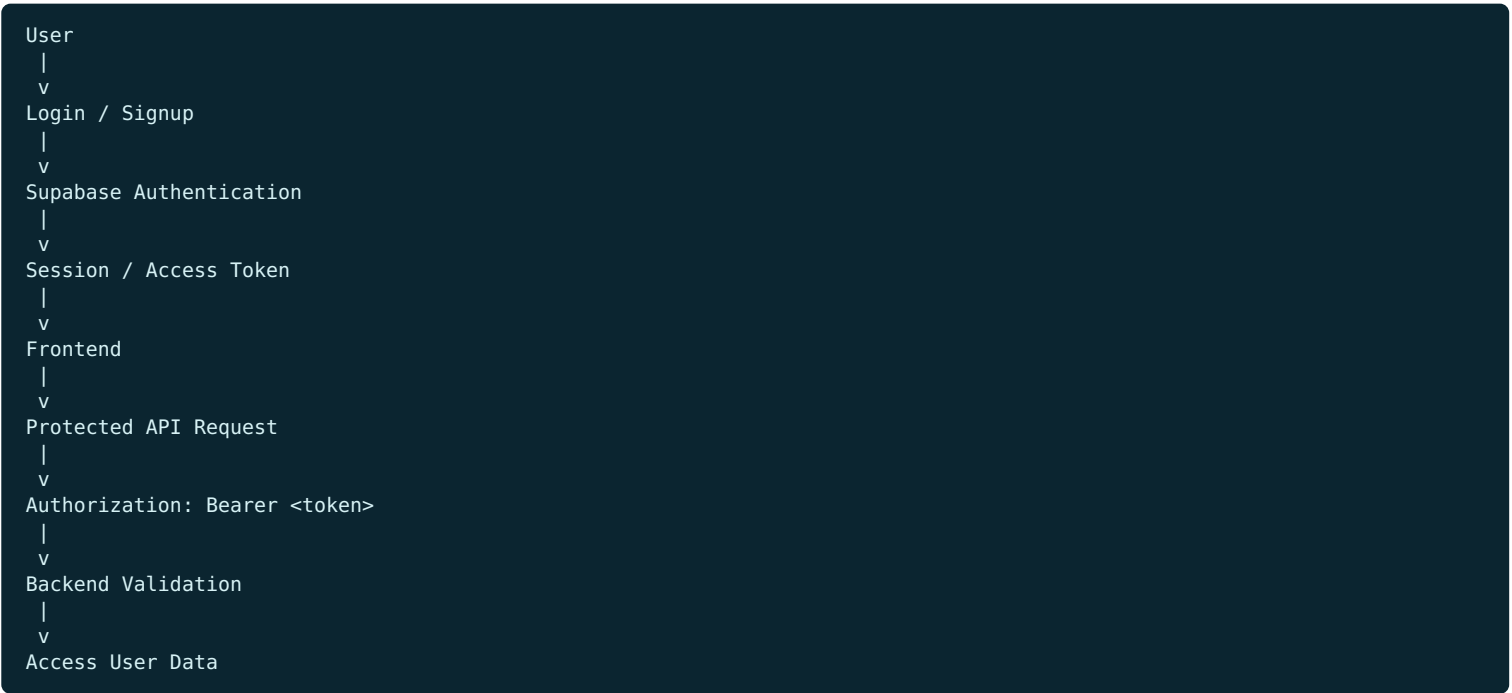
GET /readings
POST /readings

GET /readings/:id
PATCH /readings/:id
DELETE /readings/:id

GET /stats
```

Protected endpoints require authentication.

20. Authentication Flow



21. API Security

Protected API requests use Bearer Token authentication.

```
Authorization: Bearer <access_token>
```

The server validates the authentication state before allowing access to protected resources. API keys and private credentials are kept on the server side and should never be exposed in the frontend.

22. Project Structure

```
src/
|
+-- assets/
|
+-- components/
|   +-- app-shell.tsx
|   +-- chat-panel.tsx
|   +-- ui/
|
+-- data/
|   +-- kb/
|
+-- integrations/
|   +-- lovable/
|   +-- supabase/
|
+-- lib/
|   +-- ai-gateway.server.ts
|   +-- api.server.ts
|   +-- articles.ts
|   +-- auth.tsx
|   +-- chat-store.ts
|   +-- health.ts
|   +-- kb.server.ts
|
+-- routes/
|   +-- api/
|   +-- auth.tsx
|   +-- chat.$threadId.tsx
|   +-- measurements.tsx
|   +-- library.index.tsx
|   +-- library.$slug.tsx
|   +-- settings.tsx
|   +-- index.tsx
|
+-- styles.css
```

23. Installation

Step 1 — Clone Repository

```
git clone <repository-url>
cd nabd
```

Step 2 — Install Dependencies

```
npm install
```

Step 3 — Configure Environment Variables

Create a `.env` file containing the required environment variables:

```
SUPABASE_URL=your_supabase_url
SUPABASE_PUBLISHABLE_KEY=your_supabase_key
LOVABLE_API_KEY=your_lovable_api_key
```

Never commit real credentials to GitHub.

Step 4 — Run the Application

24. Testing

Authentication Testing: Signup, Login, Logout, Invalid credentials, Protected routes.

Blood Pressure Testing: Valid readings, Invalid values, Missing values, Editing readings, Deleting readings, Statistics.

AI Testing: General hypertension questions, Lifestyle questions, Medication information, Emergency scenarios, Unsupported questions, Prompt injection attempts.

API Testing: Public endpoints, Protected endpoints, Unauthorized requests, Invalid request bodies, Invalid IDs.

25. AI Evaluation

Future AI evaluation can measure:

Retrieval Quality: Precision@K, Recall@K, Context relevance

Answer Quality: Faithfulness, Groundedness, Relevance, Completeness

Safety: Unsafe recommendation rate, Hallucination rate, Emergency detection accuracy, Refusal accuracy for unsafe medical requests.

26. Performance Considerations

Potential performance improvements include:

- Caching frequently requested articles.
- Optimizing database queries.
- Pagination for blood pressure readings.
- Reducing unnecessary AI requests.
- Streaming AI responses.
- Caching embeddings in a future RAG implementation.

27. Security Considerations

The project considers: Authentication, Authorization, Row Level Security, API protection, Server-side secrets, Input validation, CORS, AI safety, Prompt injection protection, Sensitive health information.

Additional security testing should be performed before using the platform in a real healthcare environment.

28. Limitations

1. The platform is an educational prototype.
2. It does not provide medical diagnosis.
3. It does not replace doctors.
4. The current knowledge-grounding implementation can be improved with a vector database.
5. AI responses still require evaluation.
6. The platform should not be considered a clinical decision-support system without additional validation and regulatory review.

29. Future Development Roadmap

Phase 1 — Current System: Authentication, Health education, Blood pressure tracking, AI assistant, REST API, Basic medical knowledge grounding.

Phase 2 — Advanced AI: Document chunking, Embeddings, Vector database, Semantic retrieval, Reranking, Better RAG evaluation.

Phase 3 — Personalization: Personalized health insights, Reading trends, Reminders, Personalized educational recommendations.

Phase 4 — Healthcare Integration: Doctor dashboard, Patient reports, Appointment management, Wearable integration, Healthcare provider workflows.

30. Conclusion

Nabd demonstrates how Artificial Intelligence can be integrated into a health education platform while maintaining a strong focus on safety, privacy, and trustworthy information.

The project combines: AI + Medical Knowledge + Health Tracking + Authentication + REST API + Database + Security.

The ultimate goal is to make hypertension education more accessible, understandable, and useful for Arabic-speaking users while providing a foundation for future AI-powered healthcare applications.

⚠ Medical Disclaimer

Nabd is an educational and monitoring platform. It does not diagnose diseases, prescribe medication, recommend medication changes, or replace professional medical care.

Users should consult qualified healthcare professionals for diagnosis, treatment, and emergency medical decisions.